

— BHARATGEN · SPEECH INTERNSHIP

வணக்கம்



Speech-to-Speech Translation

Speak one Indian language. Hear another.

Shrutam-2 → Param-2 → Sooktam-2

CHAITANYA PATIL

— THE IDEA

Your voice — in **another language.**

Speak a sentence in Hindi. About ten seconds later you hear it spoken back to you in Tamil — in a matching voice. **Voice in, voice out**, across 12 languages, any-to-any.

Hindi

Marathi

Gujarati

Tamil

Telugu

Kannada

Bengali

Malayalam

Odia

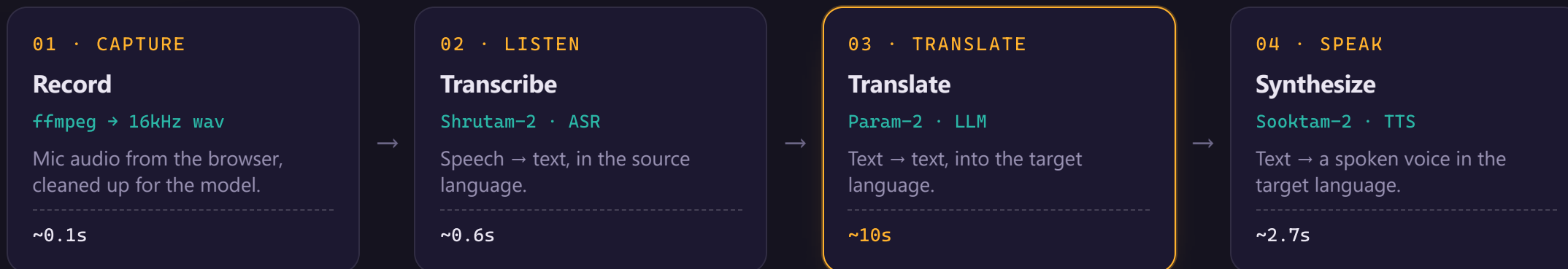
Urdu

Punjabi

Indian English

— THE PIPELINE

Four stages, mic to speaker.



Speaking and listening in the **same** language? Stage 03 is skipped entirely — the whole thing drops to ~3 seconds.

— THE TRANSLATOR

Param-2 does the **translating**.

- BharatGen's **17-billion-parameter** language model — [Param2-17B-A2.4B-Thinking](#). A "thinking" model that reasons before it answers.
- I prompt it to translate, then **strip out its private reasoning** so the TTS never reads the thoughts aloud.
- Decoding is **deterministic** — same input, same translation, every time.

model output → ~~<think>"Hello" in Tamil is the greeting vanakkam...</think>~~ வணக்கம்

————— strip the <think> block —————

sent to TTS → வணக்கம்

— ARCHITECTURE

Why translation runs **on its own**.

Param-2 needs `transformers 4.52.3`. Shrutam-2 and Sooktam-2 need `4.56.2`. They can't live in one Python process — so I split them.

Main server

:8000

ASR + TTS

`transformers 4.56.2`

↔
HTTP
9 ms

Param-2 service

:8500

translation only

`transformers 4.52.3`

The hop between them costs **9 milliseconds** — basically free. Splitting the services cost no real speed, and let each model keep the library version it needs.

— SHIPPING IT

One command stands it all up.

- Runs on a rented **NVIDIA H100** (80 GB) on RunPod — all three models loaded at once.
- `pod_setup.sh` downloads the weights, builds the environments, patches the model code, and starts every service.
- I made the setup **persistent**, so a restart reuses everything already on disk.

FIRST BOOT

~10 min

Download ~35 GB of weights, build environments from scratch.

EVERY RESTART AFTER

~30 sec

Reuse cached weights and environments — just relaunch the services.

— RESULTS

Where the **10 seconds** go.

Translate

Param-2



~10s · 67%

Synthesize

Sooktam-2



2.7s · 27%

Transcribe

Shrutam-2



0.6s · 5%

Capture

ffmpeg



0.1s · 1%

Honest numbers, from **24 profiled runs** on the H100: the median is ~10s, but the slow tail (p90) reaches **~28s**, and the very first request after boot takes **~60s** while models load.

— WHAT BROKE

Three bugs worth the story.

TTS CUT-OFF

13s → 1s

Sooktam-2 was chopping a 13-second clip down to one. **Fix:** bounded its audio-boundary search so it stops scanning past the real start.

SILENT DOWNLOAD

incomplete

The pod was missing git-lfs, so model weights downloaded **half-finished with no error**. **Fix:** switched to a verified download with an integrity check.

OUT OF MEMORY

5 GB → ok

Loading a 5 GB checkpoint blew up system RAM. **Fix:** memory-mapped the file (`mmap=True`) so it loads without copying into RAM.

— THE BOTTLENECK & WHAT'S NEXT

Param-2 is slow — so I built an **alternative.**

Translation is **two-thirds of the wait**, because Param-2 is large and stops to "think." For pure translation, a smaller specialist can be much faster.

DROP-IN

IndicTrans2

A **1-billion**-parameter translation model — no reasoning step. Built as a second service on :8501.

ZERO REWRITE

Same API

Identical endpoints to Param-2, so I switch translators with **one setting** — the rest of the pipeline doesn't change.

READY TO MEASURE

A/B test

Both can run side by side. The comparison harness is in place — next step is the head-to-head numbers.

— DEMO

Let's hear it.

A live run, end to end: speak, watch the transcript and translation appear, and play back the synthesized voice.

PLAN A

Warm it up first

Send one throwaway request before presenting, so the ~60s cold start doesn't happen on stage.

PLAN B

Recorded backup

Have a clean recording ready — insurance against the slow-tail (~28s) run on the day.

Shrutam-2 → Param-2 → Sooktam-2 · thank you